

Package ‘tisai’

April 15, 2026

Title Datasets and Utilities for AI-Enabled Time Series and Spatial Statistics

Version 0.0.1

Author Xu Liu [aut, cre]

Maintainer Xu Liu <liu.xu@sufe.edu.cn>

Description Provides datasets and helper functions used in the book
``AI Enabled Time Series and Spatial Statistics".

License GPL (>=2)

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Depends R (>= 3.5.0)

Imports dplyr,
quantmod,
zoo

LazyData true

Contents

AAPL	2
Ashare	3
CNgdp	4
CNpcgdp	5
CNpower	5
CNprice	6
DelhiWeather	7
djia	8
employ	9
Energy	10
EnergyWx	11
FANG	12
gafa_stock	13
get_Ashare_data	14
get_CNmarket_data	15
get_USmarket_data	16
GoldUSD	18
Gtemp	19

gtemp.month	19
Hare	20
HKflu	21
HousePrice	22
lap	23
Lynx	24
poverty	25
Power	26
retail	27
SCarinae	28
sunspots	29
TempAnom	29
T_econ	30
USEconomic	31
USmacro	32
walmart_sales_weekly	33
WorldMacro	35
Index	36

AAPL	<i>Daily Apple Stock Market Data</i>
------	--------------------------------------

Description

Daily Apple Inc. stock prices, trading volume, cash dividends, and split events from 2010 to 2026.

Usage

`data(AAPL)`

Format

A data frame with 4000 observations on the following 8 variables:

- `date` Trading date from April 23, 2010 to March 18, 2026.
- `open` Opening price in U.S. dollars.
- `high` Highest price of the day in U.S. dollars.
- `low` Lowest price of the day in U.S. dollars.
- `close` Closing price in U.S. dollars.
- `volume` Trading volume.
- `dividend` Cash dividend per share on dividend dates, and 0 otherwise.
- `split` Split factor on stock split dates, and 0 otherwise.

Details

This dataset is drawn from the Kaggle collection “Apple AAPL Full Stock Financial Dataset 1980–2026”. Following the user-approved trimming rule for this project, the package version keeps the latest 4000 trading-day observations, which currently span April 23, 2010 to March 18, 2026. The resulting data frame combines daily prices, trading volume, dividend records, and split events for Apple Inc. in a single object.

Source

Anadi Gupta, “Apple AAPL Full Stock Financial Dataset 1980–2026”, Kaggle. <https://www.kaggle.com/datasets/anadiskt/apple-aapl-full-stock-financial-dataset-1980-2026>

The dataset description reports underlying data sources including Yahoo Finance, Nasdaq, and public financial disclosures.

Examples

```
data(AAPL)
head(AAPL)

plot(close ~ date, data = AAPL, type = "l",
      xlab = "Date", ylab = "USD",
      main = "Apple Closing Price")
```

Ashare

Selected Chinese A-Share Stock Tickers

Description

A list of representative Chinese A-share listed companies used for constructing panel stock return data from Yahoo Finance.

Usage

```
data(Ashare)
```

Format

A data frame containing stock tickers of selected A-share companies:

`ticker` Yahoo Finance ticker symbol.

`name` Company name.

`market` Exchange code: SSE or SZSE.

Source

Yahoo Finance symbol convention for Chinese A-share listed companies: <https://finance.yahoo.com/>

Examples

```
## Not run:
data(Ashare)

## download panel data for selected A-share stocks
panel <- get_Ashare_data(
  tickers = Ashare$ticker[1:5],
  from = "2020-01-01",
  to = "2024-12-31"
)
```

```
head(panel)

## End(Not run)
```

CNgdp

China Quarterly GDP and Sectoral Value Added

Description

Quarterly measures of China's gross domestic product (GDP), including both current-price (nominal) and constant-price (real) series, together with value added of the primary, secondary, and tertiary industries.

Usage

```
data(CNgdp)
```

Format

A data frame with quarterly observations on:

date First day of the quarter (YYYY-MM-DD).
gdp GDP, current price (billion CNY).
gdp1 Primary industry value added, current price (billion CNY).
gdp2 Secondary industry value added, current price (billion CNY).
gdp3 Tertiary industry value added, current price (billion CNY).
rgdp GDP, constant price (billion CNY).
rgdp1 Primary industry value added, constant price (billion CNY).
rgdp2 Secondary industry value added, constant price (billion CNY).
rgdp3 Tertiary industry value added, constant price (billion CNY).

Source

National Bureau of Statistics of China (NBS), Quarterly Data Portal: <https://data.stats.gov.cn/>

CNpcgdp*Provincial Per Capita GDP in China, 2016–2024*

Description

Annual per capita GDP for 31 provincial-level regions in China from 2016 to 2024.

Usage

```
data(CNpcgdp)
```

Format

A data frame with 31 observations on the following 10 variables:

province Provincial-level region name.

pcgdp2016 Per capita GDP in 2016 (CNY per person).

pcgdp2017 Per capita GDP in 2017 (CNY per person).

pcgdp2018 Per capita GDP in 2018 (CNY per person).

pcgdp2019 Per capita GDP in 2019 (CNY per person).

pcgdp2020 Per capita GDP in 2020 (CNY per person).

pcgdp2021 Per capita GDP in 2021 (CNY per person).

pcgdp2022 Per capita GDP in 2022 (CNY per person).

pcgdp2023 Per capita GDP in 2023 (CNY per person).

pcgdp2024 Per capita GDP in 2024 (CNY per person).

Source

National Bureau of Statistics of China (NBS), data portal: <https://data.stats.gov.cn/>

CNpower*China Monthly Electricity Generation by Source*

Description

Monthly electricity generation in China, including total generation and generation by thermal, hydro, nuclear, wind, and solar sources.

Usage

```
data(CNpower)
```

Format

A data frame with monthly observations on:

date First day of the month (YYYY-MM-DD).
 power Total electricity generation (100 million kWh).
 thermal Thermal power generation (100 million kWh).
 hydro Hydropower generation (100 million kWh).
 nuclear Nuclear power generation (100 million kWh).
 wind Wind power generation (100 million kWh).
 solar Solar power generation (100 million kWh).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: <https://data.stats.gov.cn/>

CNprice	<i>China Monthly Consumer Price Indices (YoY)</i>
---------	---

Description

Monthly consumer price indices (CPI), reported as year-over-year indices (same month of previous year = 100), for the national aggregate as well as urban and rural residents.

Usage

```
data(CNprice)
```

Format

A data frame with monthly observations on:

date First day of the month (YYYY-MM-DD).
 cpi National CPI (YoY index, same month last year = 100).
 cpi_city Urban CPI (YoY index, same month last year = 100).
 cpi_rural Rural CPI (YoY index, same month last year = 100).

Source

National Bureau of Statistics of China (NBS), Monthly Data Portal: <https://data.stats.gov.cn/>

DelhiWeather*Daily Weather and Air Quality Data for Selected Delhi Sites*

Description

Daily weather and air-quality summaries for six monitoring locations in Delhi, India, during 2025.

Usage

```
data(DelhiWeather)
```

Format

A data frame with 2190 observations on the following 15 variables:

location Monitoring location in Delhi.

date Observation date in 2025.

lat Latitude of the monitoring location.

lon Longitude of the monitoring location.

temperature Daily mean temperature in degrees Celsius.

humidity Daily mean relative humidity percentage.

pressure Daily mean atmospheric pressure in millibars.

windspeed Daily mean wind speed in kilometers per hour.

condition Most frequent hourly weather condition recorded on that day.

aqi Daily mean air quality index.

pm25 Daily mean PM2.5 concentration in micrograms per cubic meter.

pm10 Daily mean PM10 concentration in micrograms per cubic meter.

co Daily mean carbon monoxide concentration in micrograms per cubic meter.

no2 Daily mean nitrogen dioxide concentration in micrograms per cubic meter.

obs_hours Number of hourly observations used in the daily summary.

Details

The original file contains 52560 hourly observations for six Delhi monitoring sites from January 1, 2025 to December 31, 2025. For teaching and package use, the hourly records are aggregated to a daily location-level panel by averaging the numeric indicators and retaining the most frequent hourly weather condition for each site-day.

Source

Sohail Shaikh, “Delhi Weather and AQI Dataset (2025)”, Kaggle. <https://www.kaggle.com/datasets/sohails07/delhi-weather-and-aqi-dataset-2025>

According to the dataset description, the hourly records were aggregated from real-time weather and air-quality monitoring APIs for the Delhi NCR. The collector code is available at <https://github.com/Sohail-Shaikh-07/AI-Agents/tree/main/weather-ai-agent>.

Examples

```
data(DelhiWeather)
head(DelhiWeather)

plot(aqi ~ date,
      data = subset(DelhiWeather, location == "Anand Vihar"),
      type = "l",
      xlab = "Date",
      ylab = "Daily mean AQI")
```

`djia`

Daily Dow Jones Industrial Average Prices

Description

Daily Dow Jones Industrial Average trading data redistributed from the **astsa** package and reorganized for use in **tisai**.

Usage

```
data(djia)
```

Format

A data frame with 2518 observations on the following 6 variables:

`date` Trading date from April 20, 2006 to April 20, 2016.

`open` Opening index level.

`high` Highest index level of the trading day.

`low` Lowest index level of the trading day.

`close` Closing index level.

`volume` Trading volume.

Details

The original **astsa** dataset stores these values as an `xts` object. In **tisai**, the data are reorganized as a base data frame with an explicit date column so they can be used more easily in teaching examples and package demonstrations.

Source

Redistributed from the **astsa** package. <https://cran.r-project.org/package=astsa> <https://cran.r-project.org/web/packages/astsa/astsa.pdf>

The original **astsa** documentation notes that the market data were obtained from Yahoo Finance via R financial-data tools.

Examples

```
data(djia)
head(djia)

plot(djia$date, djia$close, type = "l",
      xlab = "Date", ylab = "Index level",
      main = "DJIA Close")
```

employ

*Annual Global Employment and Unemployment Rates***Description**

Annual country-level employment and unemployment rates for the total population aged 15 and above from 2016 to 2025.

Usage

```
data(employ)
```

Format

A data frame with 1816 observations on the following 5 variables:

`iso_code` Three-letter country or territory code used by the source database.

`country` Country or territory name.

`year` Calendar year from 2016 to 2025.

`employ` Employment rate, measured as the percentage of employed persons in the working-age population.

`unemp` Unemployment rate, measured as the percentage of unemployed persons in the labour force.

Details

This dataset is drawn from a Kaggle collection based on ILOSTAT modelled estimates published by the International Labour Organization. The package version is organized as a country-year panel that combines employment and unemployment rates using exact matches on `iso_code`, `country`, `sex`, `age`, and `year`. It keeps the total population aged 15 and above from 2016 to 2025, yielding 1816 observations for 183 countries and territories.

The employment and unemployment rates in this dataset should not be interpreted as complements that add up to 100, because they use different denominators in the source definitions.

Source

Luca Lullo, “Global Employment and Unemployment Rates 1991–2025”, Kaggle. <https://www.kaggle.com/datasets/lucalullo/global-employment-unemployment-rates-1991-2025>

Underlying data source: International Labour Organization (ILO), ILOSTAT database. <https://ilostat ilo.org/>

Examples

```
data(employ)
head(employ)

china <- subset(employ, country == "China")
plot(china$year, china$employ, type = "b",
      ylim = range(c(china$employ, china$unemp)),
      xlab = "Year", ylab = "Percent",
      main = "Employment and Unemployment Rates in China")
lines(china$year, china$unemp, type = "b", col = "red")
legend("topright", legend = c("Employment", "Unemployment"),
      col = c("black", "red"), lty = 1, pch = 1, bty = "n")
```

Energy

Selected Global Energy Indicators

Description

Annual energy-related indicators for selected countries.

Usage

```
data(Energy)
```

Format

A data frame containing:

country Country name.

year Year.

pop Population.

gdp Gross domestic product.

energy Primary energy consumption.

elec Electricity generation.

renew Share of renewable energy.

Source

Our World in Data energy dataset. <https://github.com/owid/energy-data>

EnergyWx

*Daily Spanish Energy Indicators with Weather for Five Major Cities***Description**

Daily Spanish electricity demand, generation, and price indicators combined with city-level weather summaries for five major Spanish cities during 2017 and 2018.

Usage

```
data(EnergyWx)
```

Format

A data frame with 3650 observations on the following 19 variables:

`city` City name: Barcelona, Bilbao, Madrid, Seville, or Valencia.

`date` Observation date from January 1, 2017 to December 31, 2018.

`load` Daily mean actual total electricity load in megawatts.

`loadf` Daily mean forecast total electricity load in megawatts.

`price` Daily mean actual electricity price in euros per megawatt-hour.

`priceda` Daily mean day-ahead electricity price in euros per megawatt-hour.

`gas` Daily mean electricity generation from fossil gas in megawatts.

`hydro` Daily mean hydro reservoir generation in megawatts.

`nuclear` Daily mean nuclear generation in megawatts.

`solar` Daily mean solar generation in megawatts.

`wind` Daily mean onshore wind generation in megawatts.

`temp` Daily mean temperature in degrees Celsius.

`humidity` Daily mean relative humidity percentage.

`pressure` Daily mean atmospheric pressure in hectopascals.

`windspeed` Daily mean wind speed in meters per second.

`rain` Daily accumulated hourly rainfall in millimeters.

`clouds` Daily mean cloud-cover percentage.

`weather` Most frequent hourly weather category recorded on that day.

`obs_hours` Number of distinct hourly weather observations used in the daily summary.

Details

The raw data contain four years of hourly Spanish electricity variables and hourly weather observations for five cities. To satisfy the project size rule while retaining all five cities, this package version keeps the user-approved period from January 1, 2017 to December 31, 2018 and aggregates the hourly data to daily frequency. City-level weather records are first deduplicated at the city-hour level before daily summaries are computed. The energy variables are national daily indicators for Spain and therefore repeat across cities on the same date, while the weather variables vary by city.

Source

Nicholas J. H., “Hourly Energy Demand Generation and Weather”, Kaggle. <https://www.kaggle.com/datasets/nicholasjhana/energy-consumption-generation-prices-and-weather/data>

According to the dataset description, electricity demand and generation data were obtained from ENTSOE, electricity prices from Red Electrica de Espana (REE), and weather data from the Open-Weather API for the five largest cities in Spain.

Examples

```
data(EnergyWx)
head(EnergyWx)

plot(load ~ temp,
      data = subset(EnergyWx, city == "Madrid"),
      xlab = "Daily mean temperature (C)",
      ylab = "Daily mean load")
```

FANG

Daily Prices for Four FANG Stocks

Description

Daily stock prices for four FANG stocks redistributed from the **timetk** package and reorganized for use in **tisai**.

Usage

```
data(FANG)
```

Format

A data frame with 4032 observations on the following 8 variables:

symbol Stock ticker symbol: AMZN, FB, GOOG, or NFLX.
date Trading date from January 2, 2013 to December 30, 2016.
open Opening stock price in U.S. dollars.
high Highest stock price of the trading day in U.S. dollars.
low Lowest stock price of the trading day in U.S. dollars.
close Closing stock price in U.S. dollars.
volume Number of shares traded.
adjusted Adjusted closing stock price in U.S. dollars.

Details

The **tisai** version keeps the full sample provided by **timetk** and stores it as a base data frame that is ready for multi-asset time-series exercises, return calculations, and panel-style stock analyses.

Source

Redistributed from the **timetk** package. <https://cran.r-project.org/package=timetk> <https://cran.r-project.org/web/packages/timetk/timetk.pdf>

Examples

```
data(FANG)
head(FANG)

subset(FANG, symbol == "AMZN")[1:6, ]
```

gafa_stock

*Daily Prices for Four GAFA Stocks***Description**

Daily stock prices for four GAFA stocks redistributed from the **fpp3** package, where the dataset is made available through **tsibbledata**, and reorganized for use in **tisai**.

Usage

```
data(gafa_stock)
```

Format

A data frame with 5032 observations on the following 8 variables:

symbol Stock ticker symbol: AAPL, AMZN, FB, or GOOG.
 date Trading date from January 2, 2014 to December 31, 2018.
 open Opening stock price in U.S. dollars.
 high Highest stock price of the trading day in U.S. dollars.
 low Lowest stock price of the trading day in U.S. dollars.
 close Closing stock price in U.S. dollars.
 volume Number of shares traded.
 adjusted Adjusted closing stock price in U.S. dollars.

Details

In the **fpp3** ecosystem, **gafa_stock** is distributed as a **tsibble** on irregular trading days. The **tisai** version keeps the full sample and reorganizes it as a base data frame so it can be used directly in teaching examples on multi-asset prices, returns, and panel-style financial analysis.

Source

Available via the **fpp3** package, with the dataset object provided by **tsibbledata**. <https://cran.r-project.org/package=fpp3> <https://cran.r-project.org/web/packages/fpp3/fpp3.pdf> <https://cran.r-project.org/package=tsibbledata>

The original **tsibbledata** documentation states that the underlying source is Yahoo Finance historical data.

Examples

```
data(gafa_stock)
head(gafa_stock)

subset(gafa_stock, symbol == "AAPL")[1:6, ]
```

get_Ashare_data	<i>Download A-share panel data from Yahoo Finance</i>
-----------------	---

Description

This function downloads adjusted daily closing prices for selected A-share stocks from Yahoo Finance and constructs a long-format panel data set with daily log returns.

Usage

```
get_Ashare_data(
  tickers = NULL,
  from = "2018-01-01",
  to = "2025-12-31",
  save_path = NULL,
  ticker_info = NULL
)
```

Arguments

tickers	A character vector of Yahoo Finance ticker symbols, such as <code>c("600000.SS", "600036.SS", "000001.SZ")</code> . If <code>NULL</code> , the function will try to use the built-in ticker table.
from	Start date in "YYYY-MM-DD" format. Default is "2018-01-01".
to	End date in "YYYY-MM-DD" format. Default is "2025-12-31".
save_path	Optional file path for saving the resulting object as an <code>.rda</code> file. Default is <code>NULL</code> .
ticker_info	Optional data frame containing ticker information. It should include at least a column named <code>ticker</code> , and may also include <code>name</code> and <code>market</code> .

Details

If `tickers` is not provided, the function tries to use the built-in `Ashare` object (or `Ashare_tickers`, if available). Downloaded series are returned in a single long-format data frame sorted by ticker and date.

Value

A long-format data frame with columns:

- date** Trading date.
- ticker** Yahoo Finance ticker symbol.
- name** Company name, if available.
- market** Exchange code ("SSE" or "SZSE"), if available.
- close_adj** Adjusted closing price.
- ret** Daily log return.

Examples

```
## Not run:
data(Ashare)

panel <- get_Ashare_data(
  tickers = Ashare$ticker[1:5],
  from = "2020-01-01",
  to = "2024-12-31"
)

head(panel)

## End(Not run)
```

get_CNmarket_data	<i>Download Chinese financial market data from Yahoo Finance</i>
-------------------	--

Description

This function downloads daily adjusted closing prices for major Chinese financial market indices and related financial variables from Yahoo Finance, aligns trading dates across markets, and returns a data frame.

Usage

```
get_CNmarket_data(from = "2018-01-01", to = "2025-12-31", save_path = NULL)
```

Arguments

from	A character string specifying the start date in "YYYY-MM-DD" format. The default is "2018-01-01".
to	A character string specifying the end date in "YYYY-MM-DD" format. The default is "2025-12-31".
save_path	An optional character string giving the file path to save the resulting data object as an .rda file. If NULL, the data are not saved to disk.

Details

The default series include the Shanghai Composite Index, Shenzhen Component Index, CSI 300 Index, Hang Seng Index, and the USD/CNY exchange rate.

The function retrieves market data from Yahoo Finance via `quantmod::getSymbols()`. Since different markets may have different trading calendars, the downloaded series are aligned by date using inner joins, so that only common trading days are retained.

The default symbols are:

shanghai Shanghai Composite Index (000001.SS)
shenzhen Shenzhen Component Index (399001.SZ)
hs300 CSI 300 Index (000300.SS)
hsi Hang Seng Index (^HSI)
usd_cny USD/CNY exchange rate (CNY=X)

Value

A data frame containing aligned daily adjusted closing prices for the selected Chinese and related financial market series. The first column is date, followed by the downloaded market variables.

Examples

```
## Not run:
library(tisai)

CNmarket <- get_CNmarket_data()
head(CNmarket)

CNmarket2 <- get_CNmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)

CNmarket3 <- get_CNmarket_data(
  save_path = "CNmarket.rda"
)

## End(Not run)
```

get_USmarket_data	<i>Download global financial market data from FRED</i>
-------------------	--

Description

This function downloads daily financial market series from FRED, aligns dates across selected series, and returns a synchronized data frame.

Usage

```
get_USmarket_data(
  from = "2018-01-01",
  to = as.character(Sys.Date()),
  series = NULL,
  save_path = NULL,
  na_rm = TRUE
)
```

Arguments

from	A character string specifying the start date in "YYYY-MM-DD" format. Default is "2018-01-01".
to	A character string specifying the end date in "YYYY-MM-DD" format. Default is Sys.Date().
series	A named character vector of FRED series IDs. The names will be used as output variable names. If NULL, a default set of market series is used.
save_path	An optional character string giving the file path to save the resulting object as an .rda file. If NULL, the data are not saved.

na_rm Logical. If TRUE, rows with any missing values are removed after alignment. Default is TRUE.

Details

By default, the function downloads:

sp500 S&P 500 index (SP500)

nasdaq NASDAQ Composite index (NASDAQCOM)

usd_index Nominal Broad U.S. Dollar Index (DTWEXBGS)

Users may also provide additional or alternative FRED series IDs through the `series` argument.

The function downloads each series from FRED via the public CSV endpoint: <https://fred.stlouisfed.org/graph/>

Since different series may have different release calendars or missing dates, the downloaded data are merged by date. If `na_rm = TRUE`, only dates with complete observations across all selected series are retained.

The returned object includes the following attributes:

source "FRED"

series_map Named character vector of downloaded FRED series IDs

download_date System date when the function was run

Value

A data frame containing aligned daily series. The first column is date, followed by the selected market variables.

Copyright and data source note

This function downloads data from FRED for educational and research use. FRED aggregates series from different original providers. Users should check the original source and licensing terms of each selected series before redistributing the data in books, packages, or other published materials.

Examples

```
## Not run:
USmarket <- get_USmarket_data()
head(USmarket)
```

```
USmarket2 <- get_USmarket_data(
  from = "2020-01-01",
  to = "2024-12-31"
)
```

```
USmarket3 <- get_USmarket_data(
  series = c(
    sp500      = "SP500",
    nasdaq     = "NASDAQCOM",
    usd_index  = "DTWEXBGS",
    vix        = "VIXCLS"
  )
)
```

```
USmarket4 <- get_USmarket_data(
```

```

    save_path = "USmarket.rda"
  )

  ## End(Not run)

```

GoldUSD

*Daily Gold Price Series in U.S. Dollars***Description**

Daily open, high, low, close, and trading volume data for the Gold/USD market from 2016 to 2026.

Usage

```
data(GoldUSD)
```

Format

A data frame with 2554 observations on the following 6 variables:

date Trading date from January 4, 2016 to March 3, 2026.

open Opening price in U.S. dollars.

high Highest price of the day in U.S. dollars.

low Lowest price of the day in U.S. dollars.

close Closing price in U.S. dollars.

volume Trading volume.

Details

The raw file GoldUSD.csv contains 6399 daily observations from August 30, 2000 to March 3, 2026. Following the user-approved trimming rule for this project, the package version keeps the observations from 2016 onward, which yields 2554 trading-day records. The sample starts on January 4, 2016 because that is the first trading day on or after January 1, 2016.

Source

Krupal Patel, "Gold Price Dynamics", Kaggle. <https://www.kaggle.com/datasets/krupalpatel07/gold-price-dynamics>

Examples

```

data(GoldUSD)
head(GoldUSD)

plot(close ~ date, data = GoldUSD, type = "l",
      xlab = "Date", ylab = "USD",
      main = "Gold/USD Closing Price")

```

`Gtemp`*Global Annual Temperature Anomalies (1850–2017)*

Description

This dataset provides the global annual mean temperature anomalies relative to the 1961–1990 average, sourced from the HadCRUT4 dataset.

Usage

```
data(Gtemp)
```

Format

A data frame with 168 observations on the following 2 variables:

`year` The year of observation (from 1850 to 2017).

`anomaly` Global temperature anomaly (degrees Celsius) relative to the 1961–1990 mean.

Details

The dataset reflects the long-term warming trend and potential 60-70 year periodic oscillations in global temperatures.

Source

Climatic Research Unit (CRU), University of East Anglia (HadCRUT4). <https://crudata.uea.ac.uk/cru/data/temperature/>

References

Wang, S., You, J., & Huang, T. (2022). Modelling and applications for non-stationary time series in the presence of trend and period. *Scientia Sinica Mathematica*, 52(2), 177-208.

Examples

```
data(Gtemp)
plot(Gtemp$year, Gtemp$anomaly, type = "l", main = "Global Temperature Anomalies")
```

`gtemp.month`*Global Temperature Monthly Data*

Description

Global temperature monthly data from the book "AI enabled time series and spacial statistics"

Usage

```
data(gtemp.month)
```

Format

A data frame with 12 rows (months) and 49 columns (years from 1975 to 2023). Each cell contains the average temperature for that month and year.

Details

Global temperature monthly data

This dataset contains global temperature monthly data from the book "AI enabled time series and spacial statistics".

The gtemp.month dataset provides monthly global temperature data from 1975 to 2023. Rows represent months (1-12) and columns represent years. This dataset is often used in climate change analysis and time series modeling.

Source

Package 'astsa' available at <https://nickpoison.github.io/>

Examples

```
data(gtemp.month)

# Transpose the data for plotting
gtemp_t <- t(gtemp.month)

# Plot the temperature data for January
plot(rownames(gtemp_t), gtemp_t[, 1], type = "l",
     main = "January Global Temperature",
     xlab = "Year", ylab = "Temperature",
     col = "blue")

# Calculate summary statistics for each month
apply(gtemp.month, 1, summary)
```

Hare	<i>Hare Population Data</i>
------	-----------------------------

Description

Hare population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Hare)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Hare population counts

Details

Hare population data

This dataset contains hare population data from the book "AI enabled time series and spacial statistics".

The Hare dataset provides annual population counts of hares from 1845 to 1935. This dataset is often used in time series analysis and population dynamics studies.

Source

Package 'astsa' available at <https://nickpoison.github.io/>

Examples

```
data(Hare)

# Plot the hare population data
plot(Hare, main = "Hare Population Data",
      xlab = "Year", ylab = "Population",
      col = "blue")

# Calculate summary statistics
summary(Hare)
```

HKflu

Weekly Influenza Surveillance and Environmental Data in Hong Kong

Description

This dataset contains weekly records of influenza incidence rates along with various air quality and meteorological indicators in Hong Kong from December 30, 2012, to December 22, 2019.

Usage

```
data(HKflu)
```

Format

A data frame with 365 observations on the following 14 variables:

week A numeric vector indicating the sequence of monitored weeks.

influenza Influenza intensity, measured as the number of laboratory-confirmed influenza cases per 1,000 recorded consultations.

FSP Fine Suspended Particulates (PM2.5) daily average concentration.

RSP Respirable Suspended Particulates (PM10) daily average concentration.

N02 Nitrogen Dioxide (\$NO_2\$) daily average concentration.

C0 Carbon Monoxide (CO) daily average concentration.

S02 Sulphur Dioxide (\$SO_2\$) daily average concentration.

NOX Nitrogen Oxides (\$NO_x\$) daily average concentration.

O3 Ozone (\$O_3\$) daily average concentration.

temperature Weekly average temperature in degrees Celsius.
 humidity Weekly average relative humidity percentage.
 start Start date of the surveillance week.
 end End date of the surveillance week.
 wee Week index retained from the original source file.

Source

Hong Kong Environmental Protection Department and relevant health authorities.

References

Li, J., Wang, S., & You, J. (2022). Nonparametric additive models for discrete-time series with periodic features. *Acta Mathematica Sinica, Chinese Series*, 65(1), 177-204.

Examples

```
data(HKflu)
summary(HKflu)
plot(HKflu$temperature, HKflu$influenza)
```

HousePrice

Monthly House Price Index for Selected U.S. Cities

Description

Monthly Zillow Home Value Index (ZHVI) for selected major U.S. cities.

Usage

```
data(HousePrice)
```

Format

A data frame containing:

region City name.
 state State name.
 date Month.
 price Typical home value (USD).

Source

Zillow Research Housing Data. <https://www.zillow.com/research/data/>

`lap`*Local Area Pollution (LAP) Data*

Description

LAP data from the book "AI enabled time series and spacial statistics"

Usage

```
data(lap)
```

Format

A multivariate time series (mts) object with the following characteristics:

- Time period: 1970 - 1980
- Frequency: Weekly (52 observations per year)
- Number of variables: 11
- Variables:
 - tmort: Total mortality
 - rmort: Respiratory mortality
 - cmort: Cardiovascular mortality
 - tempr: Temperature
 - rh: Relative humidity
 - co: Carbon monoxide
 - so2: Sulfur dioxide
 - no2: Nitrogen dioxide
 - hycarb: Hydrocarbons
 - o3: Ozone
 - part: Particulate matter

Details

LAP data

This dataset contains LAP (Local Area Pollution) data from the book "AI enabled time series and spacial statistics".

The lap dataset provides weekly measurements of various pollution and health indicators from 1970 to 1980. This dataset is often used in environmental health studies and time series analysis of pollution effects.

Source

Package 'astsa' available at <https://nickpoison.github.io/>

Examples

```
data(lap)

# Plot the first variable (total mortality)
plot(lap[, 1], main = "Total Mortality",
     xlab = "Time", ylab = "Mortality Rate",
     col = "red")

# Calculate correlation between temperature and ozone
cor(lap[, "tempr"], lap[, "o3"], use = "complete.obs")

# Plot multiple variables
plot(lap[, c("tempr", "o3", "co")],
     main = "Temperature, Ozone, and Carbon Monoxide")
```

Lynx

Lynx Population Data

Description

Lynx population data from the book "AI enabled time series and spacial statistics"

Usage

```
data(Lynx)
```

Format

A time series object with the following characteristics:

- Time period: 1845 - 1935
- Frequency: Annual
- Values: Lynx population counts

Details

Lynx population data

This dataset contains lynx population data from the book "AI enabled time series and spacial statistics".

The Lynx dataset provides annual population counts of lynx from 1845 to 1935. This dataset is often used in time series analysis and predator-prey relationship studies.

Source

Package 'astsa' available at <https://nickpoison.github.io/>

Examples

```
data(Lynx)

# Plot the lynx population data
plot(Lynx, main = "Lynx Population Data",
      xlab = "Year", ylab = "Population",
      col = "red")

# Calculate summary statistics
summary(Lynx)
```

poverty

*Global Poverty, Inequality, and GDP Panel Data***Description**

Country-year panel data on population, GDP, poverty, and income distribution indicators from 2005 to 2024.

Usage

```
data(poverty)
```

Format

A data frame with 3980 observations on the following 11 variables:

country Country or territory name.
 iso_code Three-letter country or territory code.
 year Calendar year from 2005 to 2024.
 pop Population.
 gdp Gross domestic product.
 gdppc GDP per capita.
 poverty Poverty rate.
 gini Gini index.
 top1 Income share of the top 1 percent.
 top10 Income share of the top 10 percent.
 bottom50 Income share of the bottom 50 percent.

Details

This dataset is drawn from the Kaggle collection “Global Economic Inequality and Poverty 1980–2024”. Following the user-approved trimming rule for this project, the package version keeps the latest full years that fit within the 4000-row limit, which currently yields a country-year panel from 2005 to 2024 with 3980 observations for 199 countries and territories.

Missing values from the source are preserved rather than imputed. In the current package version, gdp and gdppc each contain 1028 missing values, poverty and gini each contain 2507 missing values, and top1, top10, and bottom50 each contain 300 missing values.

Source

Luca Lullo, “Global Economic Inequality and Poverty 1980–2024”, Kaggle. <https://www.kaggle.com/datasets/lucalullo/global-economic-inequality-and-poverty-1980-2024>

The dataset description reports underlying data sources including Our World in Data and the World Inequality Database (WID), with poverty and inequality indicators mainly based on World Bank household survey data. <https://ourworldindata.org/> <https://wid.world/>

Examples

```
data(poverty)
head(poverty)

plot(gdppc ~ year, data = subset(poverty, country == "China"),
     type = "b", xlab = "Year", ylab = "GDP per capita",
     main = "China GDP per Capita")
```

Power	<i>Hourly Household Electricity Consumption</i>
-------	---

Description

Hourly electricity consumption of a household in 2010.

Usage

```
data(Power)
```

Format

- A data frame containing:
- time Hourly timestamp
 - power Active power (kilowatt)
 - reactive Reactive power (kilowatt)
 - voltage Voltage (volt)
 - current Current intensity (ampere)
 - sub1 Energy sub-metering 1
 - sub2 Energy sub-metering 2
 - sub3 Energy sub-metering 3

Source

UCI Machine Learning Repository. <https://archive.ics.uci.edu/ml/datasets/individual+household+electric+power+consumption>

retail

*Monthly U.S. Retail Sales, Inventories, and Selected Store Categories***Description**

Monthly U.S. retail indicators from the U.S. Census Bureau, including aggregate retail sales, inventories, inventory-sales ratio, and selected seasonally adjusted store-category sales.

Usage

```
data(retail)
```

Format

A data frame with 332 observations on the following 13 variables:

`date` Month of observation from February 1992 to September 2019.

`salesn` Retail sales excluding food services, not seasonally adjusted, in millions of dollars.

`sales` Retail sales excluding food services, seasonally adjusted, in millions of dollars.

`growthn` Monthly percent change in retail sales, not seasonally adjusted.

`growth` Monthly percent change in retail sales, seasonally adjusted.

`total` Retail and food services total sales, seasonally adjusted, in millions of dollars.

`invent` Retailers inventories, seasonally adjusted, in millions of dollars.

`ratio` Retailers inventory-to-sales ratio, seasonally adjusted.

`auto` Automotive parts, accessories, and tire stores sales, seasonally adjusted, in millions of dollars.

`furn` Furniture and home furnishings stores sales, seasonally adjusted, in millions of dollars.

`build` Building materials, garden equipment, and supplies dealers sales, seasonally adjusted, in millions of dollars.

`cloth` Clothing and clothing accessory stores sales, seasonally adjusted, in millions of dollars.

`online` Electronic shopping and mail-order houses sales, seasonally adjusted, in millions of dollars.

Details

The raw directory contains multiple monthly series downloaded from a Kaggle collection that republishes U.S. Census Bureau retail data distributed through FRED. To build a compact teaching dataset, the series are merged into a single monthly data frame covering February 1992 to September 2019. The sample starts in February 1992 because the month-over-month percent change series begin one month later than the level series.

Source

U.S. Census Bureau, Monthly Retail Trade Survey, distributed by the Federal Reserve Bank of St. Louis (FRED). <https://fred.stlouisfed.org> <https://www.census.gov/retail/index.html>

Raw files in this project were collected from the Kaggle dataset “Retail and Retailers Sales Time Series Collection”. <https://www.kaggle.com/datasets/census/retail-and-retailers-sales-time-series-collection>

Examples

```
data(retail)
head(retail)

plot(retail$date, retail$sales, type = "l",
      xlab = "Date", ylab = "Millions of dollars",
      main = "U.S. Retail Sales")
```

SCarinae

*Ten-Day Visual Magnitude Series for S Carinae***Description**

Ten-day visual magnitude series for the variable star S Carinae, reorganized for use in **tisai**.

Usage

```
data(SCarinae)
```

Format

A data frame with 1189 observations on the following 2 variables:

jd Julian day index from 2431110 to 2442990 in 10-day steps.

magnitude Visual magnitude of S Carinae. Missing 10-day intervals are recorded as NA. Smaller values indicate a brighter star.

Details

The underlying observations are visual measurements of the southern variable star S Carinae. According to the project notes, the usable sample spans JD2431110 to JD2442990, corresponding to January 21, 1944 through July 30, 1976. The released **tisai** object uses the provided 10-day series and converts placeholder zeros to NA; there are 40 such missing intervals in the current sample.

Source

The official observational source is AAVSO WebObs for S Carinae. https://apps.aavso.org/webobs/results/?star=000-BBR-212&num_results=200&page=158

Project notes also reference the time-series data page maintained by Guadalupe Huerta: <https://math.unm.edu/~ghuerta/tseries/data.html>

The released **tisai** version is built from the local processed 10-day series file supplied in this project.

Examples

```
data(SCarinae)
head(SCarinae)

plot(SCarinae$jd, SCarinae$magnitude, type = "l",
      xlab = "Julian day", ylab = "Magnitude",
      main = "S Carinae")
```

sunspots	<i>Monthly Sunspot Numbers</i>
----------	--------------------------------

Description

Monthly mean relative sunspot numbers redistributed from the base R **datasets** package.

Usage

```
data(sunspots)
```

Format

A univariate time series with 2820 monthly observations from January 1749 to December 1983.

Details

This is the classical monthly sunspot series used in many forecasting and spectral-analysis examples. The object is kept as a ts series in **tisai** so it can be used directly in time-series demonstrations.

Source

Redistributed from the base R **datasets** package.

Andrews, D. F. and Herzberg, A. M. (1985). *Data: A Collection of Problems from Many Fields for the Student and Research Worker*. Springer.

The **datasets** documentation notes that the figures were collected at the Swiss Federal Observatory, Zurich until 1960, and then at the Tokyo Astronomical Observatory.

Examples

```
data(sunspots)
plot(sunspots, main = "Monthly Sunspot Numbers")
```

TempAnom	<i>Global Surface Air Temperature Anomalies from Hansen and Lebedeff</i>
----------	--

Description

Annual global surface air temperature anomalies extracted from Hansen and Lebedeff (1987) and reorganized for use in **tisai**.

Usage

```
data(TempAnom)
```

Format

A data frame with 106 observations on the following 2 variables:

year Year of observation from 1880 to 1985.

anomaly Global surface air temperature anomaly in degrees Celsius relative to the 1951–1980 mean.

Details

This series corresponds to the annual global temperature anomaly values discussed in Shao (2011) and traced through Hall and Van Keilegom (2003) back to Table 1 of Hansen and Lebedeff (1987). It provides an early annual global temperature anomaly series based on station observations with a different reference period from the Gtemp dataset in **tisai**.

Source

Hansen, J. and Lebedeff, S. (1987). Global trends of measured surface air temperature. *Journal of Geophysical Research: Atmospheres*, 92(D11), 13345–13372.

The file in this project was prepared from the table cited through:

Hall, P. and Van Keilegom, I. (2003). Using difference-based methods for inference in nonparametric regression with time series errors. *Journal of the Royal Statistical Society: Series B*, 65(2), 443–456.

Shao, X. (2011). A simple test of changes in mean in the possible presence of long-range dependence. *Journal of Time Series Analysis*, 32(6), 598–606.

Examples

```
data(TempAnom)
head(TempAnom)

plot(TempAnom$year, TempAnom$anomaly, type = "l",
      xlab = "Year", ylab = "Temperature anomaly",
      main = "Hansen-Lebedeff Global Temperature Anomalies")
```

T_econ	<i>Regional Climate and Economic Panel Data from Kotz, Levermann, and Wenz</i>
--------	--

Description

Regional panel data on climate and economic production derived from the replication package for Kotz, Levermann, and Wenz on rainfall changes and economic production, reorganized for use in **tisai**.

Usage

```
data(T_econ)
```

Format

A data frame with 14577 observations on the following 17 variables:

country Country name.

iso Three-letter country code.

year Calendar year from 2001 to 2012.

region_id Subnational spatial unit identifier from the source package.

lgdppc Log GDP per capita in U.S. dollars.

d1gdppc Annual change in log GDP per capita.
 agdppc Agricultural GDP per capita in U.S. dollars.
 mangdppc Manufacturing GDP per capita in U.S. dollars.
 servgdppc Services GDP per capita in U.S. dollars.
 p5_totalpr Source-package precipitation indicator.
 t5_mean Source-package temperature mean indicator.
 t5_varc Source-package temperature variability indicator.
 t5_varm Source-package temperature variability indicator.
 t5_seas_diff Source-package temperature seasonality indicator.
 rich High-income country dummy from the source package.
 lon Longitude of the spatial unit centroid.
 lat Latitude of the spatial unit centroid.

Details

The original T_econ.dta file contains a larger subnational panel with many climate-derived features. For the **tisai** release, the data are organized as a continuous-year subset covering 2001–2012, which retains 14577 observations from 75 countries and 1446 spatial units while preserving the subnational structure.

To stay close to the original replication package, several climate variables retain condensed source notation such as p5_totalpr and t5_mean.

Source

Kotz, M., Levermann, A. and Wenz, L. (2022). The effect of rainfall changes on economic production. *Nature*, 601, 223–227. doi:10.1038/s41586021042838

Replication package: <https://doi.org/10.5281/zenodo.5657457>

The Zenodo record notes that the economic data are derived from the DoSE database and the climate data are derived from the ERA5 reanalysis.

Examples

```

data(T_econ)
head(T_econ)

subset(T_econ, iso == "CHN")[1:6, ]

```

USeconomic

U.S. Quarterly National Economic Output and Expenditure Components

Description

This dataset provides a structural (quarterly) snapshot of the U.S. economy from 2000.1.1 to 2025.7.1, consisting of 103 observations.

It integrates the major components of Gross Domestic Product (GDP): - Aggregate Output: Nominal and Real GDP. - Domestic Demand: Personal consumption (PCE) and Private investment. - External Sector: Real exports and imports of goods and services.

The dataset allows students to explore the nonlinear dynamics of investment shocks and the global trade balance using advanced AI structural models.

Usage

```
data(USEconomic)
```

Format

A data frame with aligned quarterly observations on the following 7 variables:

```
date  Start date of the quarter (YYYY-MM-DD).
gdp    Gross Domestic Product (Billions of Dollars).
real_gdp  Real Gross Domestic Product (Billions of Chained 2017 Dollars).
pce    Personal Consumption Expenditures (Billions of Dollars).
invest  Real Gross Private Domestic Investment (Billions of Chained 2017 Dollars).
export  Real Exports of Goods and Services (Billions of Chained 2017 Dollars).
import  Real Imports of Goods and Services (Billions of Chained 2017 Dollars).
```

Source

U.S. Bureau of Economic Analysis (BEA) via Federal Reserve Bank of St. Louis (FRED). Data IDs: GDP, GDPC1, PCE, GPDIC1, EXPGSC1, and IMPGSC1. Visit: <https://fred.stlouisfed.org/>

References

U.S. Bureau of Economic Analysis, "National Income and Product Accounts". Retrieved from FRED: <https://fred.stlouisfed.org/categories/106>

Examples

```
data(USEconomic)
# Calculate Investment-to-GDP ratio
inv_ratio <- USEconomic$invest / USEconomic$real_gdp
plot(USEconomic$date, inv_ratio, type="l", ylab="Investment/GDP Ratio")
```

USmacro

U.S. Monthly Macroeconomic Pulse: Labor, Price, and Monetary Indicators

Description

This dataset provides a high-frequency (monthly) snapshot of the U.S. economy from 2000.1.1 to 2025.10.1, consisting of 310 observations.

The variables cover four critical dimensions: 1. Labor Market: Unemployment rate and Nonfarm payrolls. 2. Price Level: Consumer Price Index (CPI) for inflation tracking. 3. Monetary Policy: Federal Funds Rate and 10-Year Treasury yields. 4. Real Activity: Industrial production and Housing starts.

This data is ideal for teaching lead-lag relationships, regime switching, and high-frequency forecasting using LSTM or Transformer architectures.

Usage

```
data(USmacro)
```

Format

A data frame with aligned monthly observations on the following 8 variables:

date First day of the month (YYYY-MM-DD).

unrate Civilian Unemployment Rate (Percent).

cpi Consumer Price Index for All Urban Consumers (Index 1982-1984=100).

interest Federal Funds Effective Rate (Percent).

indpro Industrial Production Index (Index 2017=100).

employ All Employees, Total Nonfarm (Thousands of Persons).

housing New Privately Owned Housing Units Started (Thousands of Units).

bond10y 10-Year Treasury Constant Maturity Rate (Percent).

Source

Federal Reserve Bank of St. Louis (FRED). Data accessible via the following IDs: UNRATE, CPI-AUCSL, FEDFUNDS, INDPRO, PAYEMS, HOUST, and GS10. Visit: <https://fred.stlouisfed.org/>

References

Federal Reserve Bank of St. Louis, Federal Reserve Economic Data (FRED). <https://fred.stlouisfed.org>

Examples

```
data(USmacro)
# Visualize the relationship between interest rates and housing starts
plot(USmacro$interest, USmacro$housing,
     main="Interest Rates vs Housing Starts")
```

walmart_sales_weekly	<i>Weekly Walmart Sales Sample</i>
----------------------	------------------------------------

Description

Weekly Walmart sales data redistributed from the **timetk** package and reorganized for use in **tisai**.

Usage

```
data(walmart_sales_weekly)
```

Format

A data frame with 1001 observations on the following 17 variables:

`id` Series identifier combining store and department.

`store` Store identifier.

`dept` Department identifier.

`date` Weekly date from February 5, 2010 to October 26, 2012.

`weekly_sales` Weekly department sales.

`is_holiday` Logical indicator for holiday weeks.

`type` Store type.

`size` Store size in square feet.

`temperature` Regional average temperature.

`fuel_price` Regional fuel price.

`markdown1` Promotional markdown feature 1.

`markdown2` Promotional markdown feature 2.

`markdown3` Promotional markdown feature 3.

`markdown4` Promotional markdown feature 4.

`markdown5` Promotional markdown feature 5.

`cpi` Consumer price index.

`unemployment` Regional unemployment rate.

Details

The original **timetk** dataset is a sample derived from the Kaggle competition “Walmart Recruiting - Store Sales Forecasting”. The **tisai** version keeps the same sample coverage, which contains seven department-level weekly series from Store 1, while standardizing the object as a base data frame with lowercase variable names. Missing promotional markdown values are preserved.

Source

Redistributed from the **timetk** package. <https://cran.r-project.org/package=timetk> <https://cran.r-project.org/web/packages/timetk/timetk.pdf>

In **timetk**, this dataset is described as a sample derived from the Kaggle competition “Walmart Recruiting - Store Sales Forecasting”. <https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting>

Examples

```
data(walmart_sales_weekly)
head(walmart_sales_weekly)

subset(walmart_sales_weekly, dept == 1)[1:6, ]
```

WorldMacro*Global Macroeconomic Panel Data*

Description

Panel data of macroeconomic indicators for multiple countries, including GDP per capita, population, and inflation rate.

Usage

```
data(WorldMacro)
```

Format

A data frame containing:

country Country name.

iso2c Two-letter country code.

iso3c Three-letter country code.

year Year.

gdp_pc GDP per capita (constant USD).

population Total population.

inflation Inflation rate.

Source

World Bank World Development Indicators <https://data.worldbank.org>

Index

* datasets

AAPL, [2](#)
Ashare, [3](#)
CNgdp, [4](#)
CNpcgdp, [5](#)
CNpower, [5](#)
CNprice, [6](#)
DelhiWeather, [7](#)
djia, [8](#)
employ, [9](#)
Energy, [10](#)
EnergyWx, [11](#)
FANG, [12](#)
gafa_stock, [13](#)
GoldUSD, [18](#)
Gtemp, [19](#)
gtemp.month, [19](#)
Hare, [20](#)
HKflu, [21](#)
HousePrice, [22](#)
lap, [23](#)
Lynx, [24](#)
poverty, [25](#)
Power, [26](#)
retail, [27](#)
SCarinae, [28](#)
sunspots, [29](#)
T_econ, [30](#)
TempAnom, [29](#)
USEconomic, [31](#)
USmacro, [32](#)
walmart_sales_weekly, [33](#)
WorldMacro, [35](#)

AAPL, [2](#)
Ashare, [3](#)

CNgdp, [4](#)
CNpcgdp, [5](#)
CNpower, [5](#)
CNprice, [6](#)

DelhiWeather, [7](#)
djia, [8](#)

employ, [9](#)
Energy, [10](#)
EnergyWx, [11](#)

FANG, [12](#)

gafa_stock, [13](#)
get_Ashare_data, [14](#)
get_CNmarket_data, [15](#)
get_USmarket_data, [16](#)
GoldUSD, [18](#)
Gtemp, [19](#)
gtemp.month, [19](#)

Hare, [20](#)
HKflu, [21](#)
HousePrice, [22](#)

lap, [23](#)
Lynx, [24](#)

poverty, [25](#)
Power, [26](#)

retail, [27](#)

SCarinae, [28](#)
sunspots, [29](#)

T_econ, [30](#)
TempAnom, [29](#)

USEconomic, [31](#)
USmacro, [32](#)

walmart_sales_weekly, [33](#)
WorldMacro, [35](#)